

Radiative Corrections to the J/ψ photoproduction reaction

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The J/ψ photoproduction reaction, has garnered significant attention in recent years due to the discovery of LHCb pentaquarks. Additionally, measurements of the near-threshold cross section provide valuable insights into the proton's mass radius. Several experiments at Jefferson Lab have focused on this topic. Precision measurements necessitate the evaluation of radiative corrections. These corrections have traditionally been estimated based on the Bethe-Heitler process. However, this approximation significantly deviates compared to the exact calculations presented in this paper.

I. Introduction

The J/ψ photoproduction reaction $\gamma p \rightarrow J/\psi p$ has garnered significant attention in recent years, particularly due to the discovery of LHCb pentaquarks. Moreover, measurements of the near-threshold cross section provide valuable insights into the proton's structure, including the determination of its mass radius. Several experiments at Jefferson Lab have focused on this topic. Precision measurements necessitate the evaluation of radiative corrections. These corrections have traditionally been estimated based on the Bethe-Heitler process. However, this approximation significantly deviates compared to the exact calculations presented in this paper.

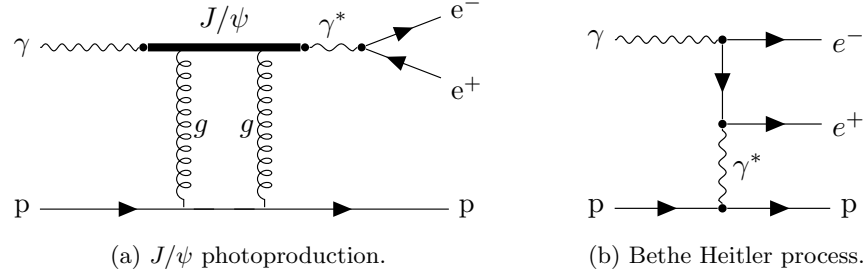


FIG. 1: Feynman diagrams illustrating the J/ψ photoproduction and the Bethe Heitler process.

Figure 1 presents the schematic Feynman diagram for the J/ψ photoproduction along with the Bethe-Heitler process. The soft-photon radiation corrections for the Bethe-Heitler process were calculated in [1]. These include vertex corrections, lepton self-energy corrections, and real photon emission from the outgoing leptons. Conversely, the J/ψ radiation corrections include only the correction to the J/ψ decay to leptons. These corrections are considered in this paper, see Figure 2.

We can conclude from these considerations that radiative corrections are expected to differ for these particular processes, even in the approximation of soft-photon emission. Numerical calculations support this conclusion.

II. Vector mesons decay to electron-positron or muon pair

In the framework of the vector-dominance model (VDM) hadron electromagnetic current reads [2]:

$$\langle V(k), \lambda | j^\mu(0) | 0 \rangle = \frac{em_V^2}{f_V} \epsilon_\lambda^\mu(V), \quad (1)$$

where e is the positron charge magnitude ($e^2 = 4\pi\alpha$), m_V is a vector meson mass, k_μ is four-momentum of a vector meson, λ is the vector meson's helicity, ϵ_λ^μ is the vector meson polarization vector and f_V is dimensionless coupling

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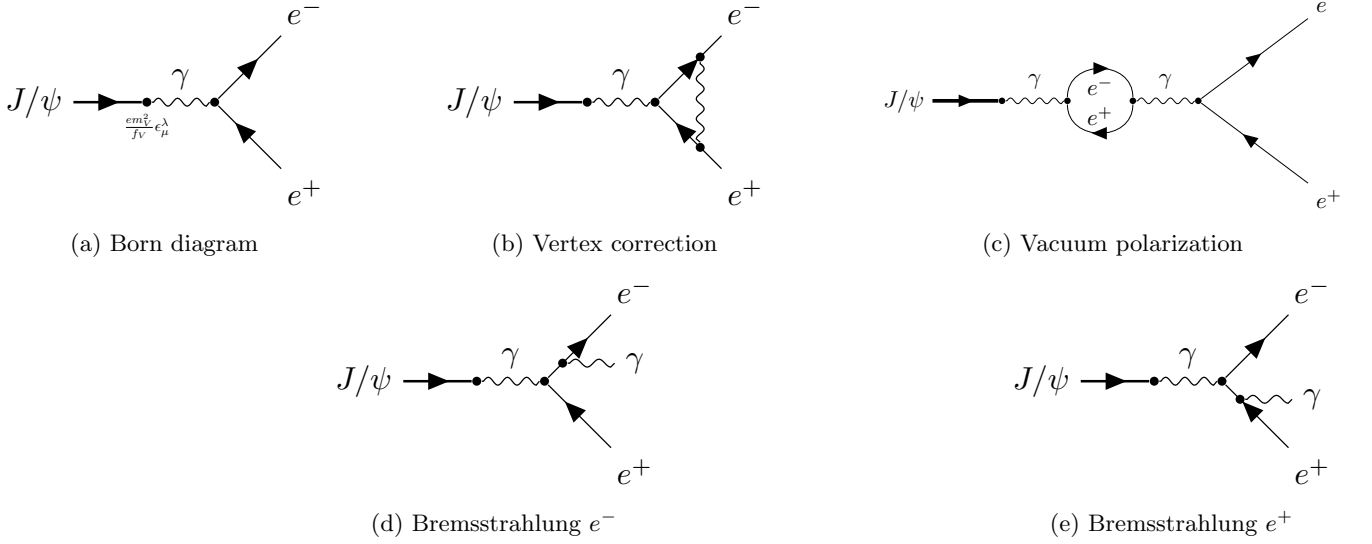


FIG. 2: Radiative corrections for the $J/\psi \rightarrow l^+l^-$ decay.

constant. The constant f_V is sometimes denote by $2\gamma_V$. The choice of mass in the numerator is arbitrary. It is conventional to choose it as a vector meson mass.

The vector meson polarization vector ϵ_μ is a four-vector that describes the polarization state of a vector meson. For a vector meson with mass m_V and four-momentum p_μ , the polarization vector satisfies the following properties:

1. Transversality: $\epsilon_\mu p_\mu = 0$. This means that the polarization vector is orthogonal to the meson's momentum vector.
2. Normalization: $\epsilon_\mu \epsilon^\mu = -1$. This normalization condition ensures that the polarization vector is properly normalized. For massive vector mesons, the polarization vector can have different states corresponding to the possible spin projections.

For a vector meson with spin along the direction $\mathbf{n}$, the polarization vector is given by $\epsilon^\mu = (0, \mathbf{n})$.

The matrix element of the decay $V \rightarrow l^+l^-$ of the Born term Figure 2(a) reads [3]:

$$\mathcal{M} = \frac{e^2 m_V^2}{f_V} \epsilon_\mu^\lambda(V) \frac{1}{k^2} \bar{u} \gamma_\mu v = \frac{e^2}{f_v} \epsilon_\mu^\lambda(V) \bar{u} \gamma_\mu v, \quad (2)$$

and

$$|\mathcal{M}|^2 = \frac{e^4}{f_V^2} \sum_{s_1, s_2} \bar{u}(p_1, s_1) \not{\epsilon}^\lambda v(p_2, s_2) \bar{v}(p_2, s_2) \not{\epsilon}^\lambda u(p_1, s_1), \quad (3)$$

where $p_1 = p_-$ and $p_2 = p_+$ are the 4-vectors of leptons and m is the lepton mass. In the rest frame of the vector meson $p_1 = (\frac{m_V}{2}, \vec{p}_l)$, $p_2 = (\frac{m_V}{2}, -\vec{p}_l)$, $|\vec{p}_l| = \frac{m_V}{2} \sqrt{1 - \frac{4m^2}{m_V^2}}$ and $\epsilon^\lambda \cdot k = m_V \epsilon_0^\lambda = 0$.

$$\begin{aligned} \frac{1}{4} \sum_{s_1, s_2} \bar{u}(p_1, s_1) \not{\epsilon}^\lambda v(p_2, s_2) \bar{v}(p_2, s_2) \not{\epsilon}^\lambda u(p_1, s_1) &= \frac{1}{4} \text{Tr}\{\not{\epsilon}(\not{p} - m) \not{\epsilon}(\not{p} + m)\} \\ &= 2(p_1 \cdot \epsilon)(p_2 \cdot \epsilon) - \epsilon^2(p_1 \cdot p_2) - m^2 \epsilon^2 = -2|\vec{p}_1|^2 \cos^2 \theta + \frac{m_V^2}{2} \\ &= \frac{m_V^2}{2} - \frac{m_V^2}{2} \left(1 - \frac{4m^2}{m_V^2}\right) \cos^2 \theta = \frac{m_V^2}{2} \left(\sin^2 \theta + \frac{4m^2}{m_V^2} \cos^2 \theta\right) \end{aligned} \quad (4)$$

Thus, we get

$$|\mathcal{M}|^2 = 2m_V^2 \frac{e^4}{f_v^2} \left(\sin^2 \theta + \frac{4m^2}{m_V^2} \cos^2 \theta \right) \quad (5)$$

The differential decay probability is defined by the equation

$$d\Gamma = \frac{1}{32\pi^2} |\mathcal{M}|^2 \frac{|p_l|}{m_V^2} d\Omega = \frac{1}{64\pi^2} \frac{|\mathcal{M}|^2}{m_V} \sqrt{1 - \frac{4m_l^2}{m_V^2}} d\Omega. \quad (6)$$

After integrating over $d\Omega = d\phi d\cos\theta$ we are getting the total partial width $\Gamma_0(V \rightarrow l^- + l^+)$:

$$\Gamma_0(V \rightarrow l^- + l^+) = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3} \left(1 + \frac{2m_l^2}{m_V^2} \right) \sqrt{1 - \frac{4m_l^2}{m_V^2}}. \quad (7)$$

In the approximation $m_l \ll m_V$ the equation simplifies to

$$\Gamma_0(V \rightarrow l^- + l^+) = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3}. \quad (8)$$

The coupling constant f_V is defined directly by the partial decay width $\Gamma_0(V \rightarrow l^+ l^-)$ in Eq. 8. The vector-dominance model constants f_V are listed in Table I.

TABLE I: Vector Dominance Model Constants

	Mass MeV	Full Width Γ_0 MeV	$BR(V \rightarrow e^+ e^-)$	$\Gamma(V \rightarrow e^+ e^-)$ MeV	f_V
$\rho(770)$	775.96	149.1	4.72e-5	7.04e-3	4.96
$\omega(782)$	782.66	8.68	7.38e-5	6.41e-4	16.5
$\phi(1020)$	1019.46	4.249	2.979e-4	1.27e-3	13.4
$J/\psi(1S)$	3096.9	0.0926	5.971e-2	5.53e-3	11.18

III. Radiation Corrections

The first-order QED radiation corrections to the width of the vector meson $V \rightarrow l^+ l^-$ decay, electrons and muons, were calculated a while ago in [4].

The total width $\Gamma(V \rightarrow l^+ l^-)$ with the radiative corrections reads

$$\Gamma = \Gamma_0(1 + \delta_S + \delta_H), \quad (9)$$

where Γ_0 is the Born term total width (Figure 2a), δ_S stands for the soft-photons radiative corrections determined by diagrams in Figures 2b-2e as well as of real photons whose energy is less a cut-off energy $E_\gamma < \varepsilon$, and δ_H accounts for the emission of the hard bremsstrahlung photons with the energy $E_\gamma > \varepsilon$. The corrections to the total width Γ define actually radiative corrections for the cross section of the photoproduction process $\gamma p \rightarrow J/\psi p \rightarrow l^+ l^- p$

$$\sigma = \sigma_0(1 + \delta_S + \delta_H). \quad (10)$$

It is important to understand that all the above-mentioned amplitudes, except Born and vac, contain infrared-divergent terms (tending to infinity in the limit of very soft, or ‘infrared’, photons). These are canceled out completely in the sum of all diagrams in Figure 2(b)-(e), which is therefore finite. More precisely, there are the following cancellations of the infrared divergences: between the lepton vertex correction and the lepton bremsstrahlung correction. The vacuum polarization corrections, shown in Figure 2(c), is finite. It is convenient to separate the amplitude into ‘soft’ and ‘hard’ parts, The label ‘soft’ implies the assumption that at least one virtual photon is soft. Infrared divergences are completely contained in the soft part.

It is important to understand that first-order bremsstrahlung is taken into account in both cases, and the separation between these two types of events is based on the energy of the emitted photon. If this energy does not exceed the

cut-off value $E_\gamma^{cut} < \varepsilon$ (which can be set to, for example, 1 MeV for the experiments of interest), then the process can be effectively considered as pure $\gamma p \rightarrow J/\psi p \rightarrow l^+ l^- p$ without bremsstrahlung photons since the experiment can not distinguish it from the $\gamma p \rightarrow J/\psi p \rightarrow l^+ l^- p \gamma$ process. In this case, it is not necessary to consider such soft photons in the Monte Carlo simulation, so we can analytically integrate the cross section of the first-order bremsstrahlung $\gamma p \rightarrow J/\psi p \rightarrow l^+ l^- p \gamma$ process over all directions of the emitted photon and over its energy in the range $E_\gamma < \varepsilon$. Moreover, we can confidently use the primary soft-photon approximation to perform this integration, because it works well for very low-energy photons.

The quantity $\delta = \delta_S + \delta_H$ includes both the corrections due to the emission of a real bremsstrahlung photon and the virtual-photon corrections. However, corrections of both these types are infrared-divergent, so that only the total correction δ can be defined uniquely. The only exception is the vacuum polarization correction δ_{vacuum} , which is finite and therefore can be determined individually.

IV. Soft Radiation Corrections

In any experiment with finite energy resolution, the emission of additional soft photons with $|k| < \varepsilon$ is undetectable and thus gets subsumed in the cross section for the elastic scattering process. The integrals over a (small) momentum sphere $|k| < \varepsilon$ are shown in Eq. 11 for electrons and in Eq. 12 for muons [4]:

$$\delta_S(M_V, \varepsilon) = -\frac{4\alpha}{\pi} \left[\left(\ln \frac{M_V}{m_e} - \frac{1}{2} \right) \left(\ln \frac{M_V}{2\varepsilon} - \frac{13}{12} \right) + \frac{17}{72} - \frac{\pi^2}{12} \right]. \quad (11)$$

$$\delta_S(M_V, \varepsilon) = -\frac{4\alpha}{\pi} \left[\left(\ln \frac{M_V}{m_\mu} - \frac{1}{2} \right) \left(\ln \frac{M_V}{2\varepsilon} - \frac{13}{12} \right) + \frac{17}{72} - \frac{\pi^2}{12} + \left(\frac{5}{18} - \frac{1}{3} \ln \frac{M_V}{m_e} \right) \right]. \quad (12)$$

Soft photon limit is defined by a scaling of the momenta of virtual photons in the loops and real photons momenta in the bremsstrahlung process, with respect to external scales.

Soft radiative corrections δ_S for electrons and muons as a function ε are presented in Figure 3.

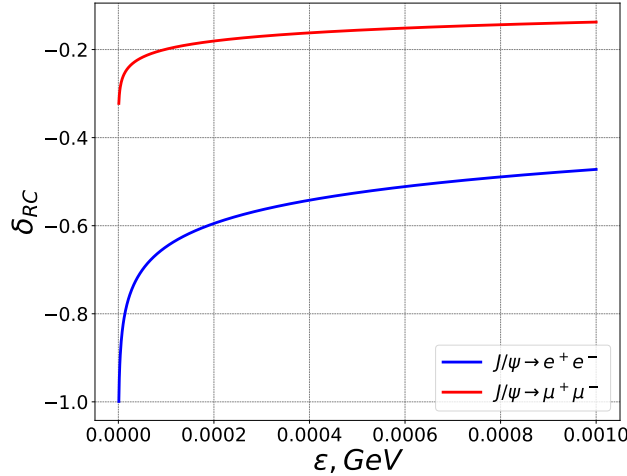


FIG. 3: Soft radiative corrections $\delta_S(M_{J/\psi}, \varepsilon)$ as a function of ε for the $J/\psi \rightarrow e^+e^-$ and $J/\psi \rightarrow \mu^+\mu^-$ decay modes.

V. Hard Radiation Corrections

The exact calculation of the hard radiative corrections δ_H would require a rather complicated angular integration of the decay probability. At high energy it could be done in the peaking approximation when additional hard photon

is emitted parallel to the lepton (or antilepton) momentum. According to [5] and [6] we get

$$\frac{d\Gamma_H}{dk} = \Gamma_0 \frac{2\alpha}{\pi} \left[2 \ln \left(\frac{M_V}{m_e} \sqrt{1 - \frac{2k}{M_V}} \right) - 1 \right] \frac{1}{k} \left(1 - \frac{2k}{M_V} + \frac{2k^2}{M_V^2} \right). \quad (13)$$

The integration of Eq. 13 in the range of the photon energy (ε, k_{max}) is presented in Eq. 14 for electrons and Eq. 15 for muons.

$$\delta_H(M_V, \varepsilon, k_{max}) = -\frac{4\alpha}{\pi} \left[\left(\ln \frac{M_V}{m_e} - \frac{1}{2} \right) \left(\ln \frac{\varepsilon}{k_{max}} + \frac{3}{4} \right) + \frac{1}{2} L_2(r_{max}) + R(r_{max}) + \frac{r_{max}^2}{16} - \frac{3}{8} r_{max} \right] \quad (14)$$

$$\delta_H(M_V, \varepsilon, k_{max}) = -\frac{4\alpha}{\pi} \left[\left(\ln \frac{M_V}{m_\mu} - \frac{1}{2} \right) \left(\ln \frac{\varepsilon}{k_{max}} + \frac{3}{4} \right) + \frac{1}{2} L_2(r_{max}) + R(r_{max}) + \frac{r_{max}^2}{16} - \frac{3}{8} r_{max} \right] \quad (15)$$

$$r_{max} = \frac{2k_{max}}{M_V} \leq 1 - \frac{4m_l^2}{M_V^2} \equiv \beta^2 \quad (16)$$

$$L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt \quad (17)$$

The exact one-loop calculation of the same decay width is given in Ref. [7], which uses a different but equivalent set of variables. The correspondence is: the soft-hard separator ε used here is denoted ω there; the maximum hard-photon energy k_{max} is denoted ω_{cut} ; and $r_{max} = 2k_{max}/M_V$ is denoted $r_{cut} = \omega_{cut}/E_1$ with $E_1 = M_V/2$, so that $r_{cut} \equiv r_{max}$. The large logarithm used there, $L = \ln(s/m_l^2)$ with $s = M_V^2$, is twice the one used here, $L = 2 \ln(M_V/m_l)$.

$$R(r_{max}) = \left(r_{max} - \frac{r_{max}^2}{4} - \frac{3}{4} \right) \left[\ln \left(\frac{M_V}{m_e} \sqrt{1 - r_{max}} \right) - \frac{1}{2} \right], \quad (18)$$

$$R(r_{max} = 1) = 0, \quad L_2(r_{max} = 1) = \frac{\pi^2}{6}$$

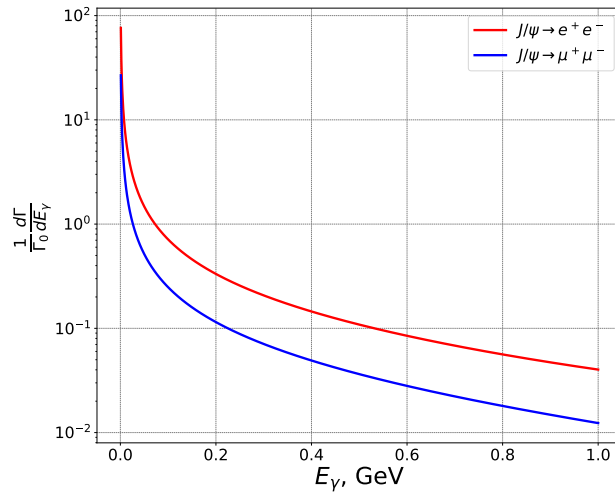


FIG. 4: Hard photon energy distribution $\frac{1}{\Gamma_0} \frac{d\Gamma}{dE_\gamma}$ for the $J/\psi \rightarrow e^+e^-$ and $J/\psi \rightarrow \mu^+\mu^-$ decay modes.

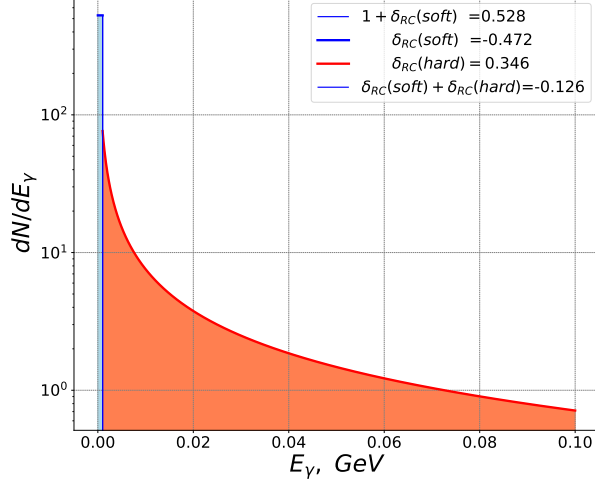


FIG. 5: Radiative photon energy distribution for $\varepsilon=0.001$ GeV and $k_{max}=0.1$ GeV. The calculation is done for $J/\psi \rightarrow e^+e^-$ decay.

The k_{max} photon energy depends on the experimental cut off energy. Its kinematic maximum is reached when the lepton pair is produced with the smallest possible invariant mass, $M_{l+l-} = 2m_l$, which gives

$$k_{max}^{\max} = \frac{M_V^2 - 4m_l^2}{2M_V} = \frac{M_V}{2} - \frac{2m_l^2}{M_V}, \quad \text{so that} \quad r_{max} \leq 1 - \frac{4m_l^2}{M_V^2} \equiv \beta^2,$$

which is the bound quoted in Eq. 16, and agrees with Eq. 21 evaluated at $M_{e^+e^-} = 2m_l$. For the electron channel $\beta^2 = 1 - 1.1 \times 10^{-7}$, so $k_{max}^{\max} = M_V/2$ to any relevant accuracy; for muons $\beta^2 = 0.99535$ and the massless endpoint lies outside the physical region. Hard photon energy distributions for electrons and muons are presented in Figure 4.

The energy distribution for $\varepsilon=0.001$ GeV and $k_{max}=0.1$ GeV parameter set is presented in Figure 5. The blue curve is for soft photons $E\gamma < \varepsilon$ and red curve for hard photons $\varepsilon < E\gamma < k_{max}$.

Hard radiative corrections $\delta_H(M_{J/\psi}, \varepsilon, k_{max})$ as a function of k_{max} with $\varepsilon=0.001$ GeV is presented in Figure 6.

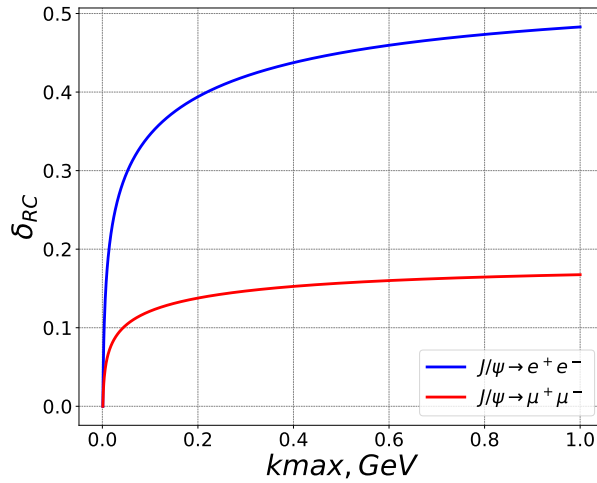


FIG. 6: Hard radiative corrections $\delta_H(M_{J/\psi}, \varepsilon, k_{max})$ as a function of k_{max} with $\varepsilon=0.001$ GeV, for the $J/\psi \rightarrow e^+e^-$ and $J/\psi \rightarrow \mu^+\mu^-$ decay modes.

VI. Total Radiation Corrections

The total radiative corrections, $\delta(k_{max}) = \delta_S(M_V, \varepsilon) + \delta_H(M_V, \varepsilon, k_{max})$ does not depend on the ε any more. Eq. 19 presents total corrections for electrons and Eq. 20 for muons

$$\delta(M_V, k_{max}) = \delta_S + \delta_H = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{M_V}{m_e} \right) \left(\ln r_{max} + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r_{max}) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r_{max}) + \frac{r_{max}^2}{16} - \frac{3}{8} r_{max} \right] \quad (19)$$

$$\delta(M_V, k_{max}) = \delta_S + \delta_H = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{M_V}{m_\mu} \right) \left(\ln r_{max} + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r_{max}) - \frac{\pi^2}{6} \right) + \frac{17}{72} + \left(\frac{5}{18} - \frac{1}{3} \ln \frac{M_V}{m_e} \right) + R(r_{max}) + \frac{r_{max}^2}{16} - \frac{3}{8} r_{max} \right] \quad (20)$$

Soft, hard and total radiative corrections as a function of the vector meson mass ($\varepsilon=0.001$ GeV, $k_{max}=0.1$ GeV) are presented in Figure 7.

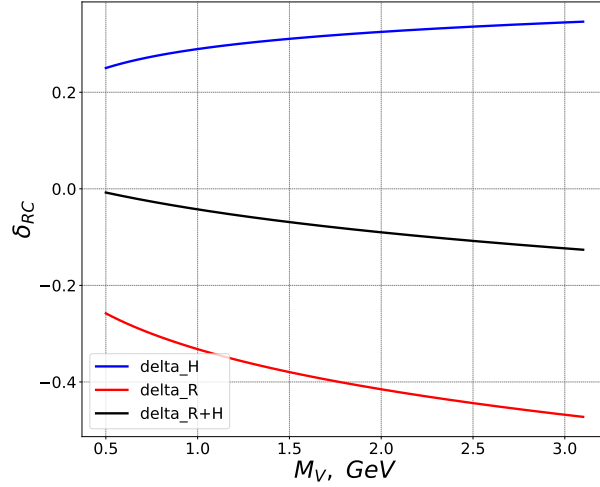


FIG. 7: Soft, hard and total radiative corrections as a function of the vector meson mass, $\varepsilon=0.001$ GeV and $k_{max}=0.1$ GeV.

Total radiative corrections as a function of $r_{max} = 2k_{max}/M_V$ for electrons and muons are presented in Figure 8, each curve covering the full physical range of its channel, $r_{max} \leq \beta^2$. In this variable the result may be compared directly with Fig. 4 of Ref. [7].

VII. Comparison of the Soft Radiation Corrections for the Vector mesons photoproduction and Bethe-Heitler processes

Comparison of the soft radiative corrections for Bethe Heitler (BH) process [1] with $M_{e^+e^-} = M_{J/\psi}$ and J/ψ photoproduction [4] is presented in Figure 9. At typical cut-off soft photons energy $\varepsilon = 0.1$ GeV $\delta_{RC}(BH) = -20\%$ and $\delta_{RC}(J/\psi) = -12\%$.

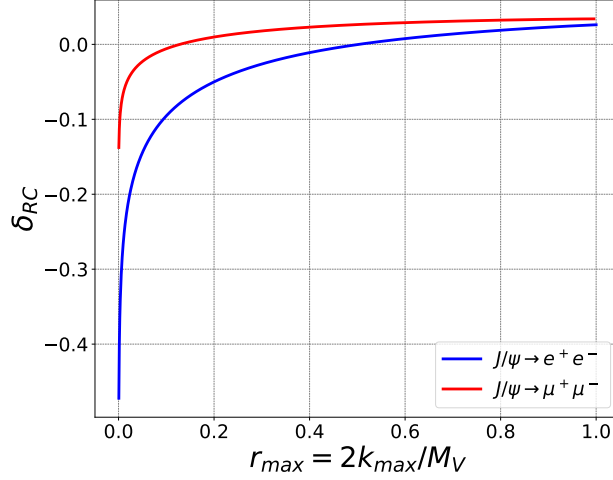


FIG. 8: Total radiative corrections as a function of $r_{max} = 2k_{max}/M_V$ for the $J/\psi \rightarrow e^+e^-$ and $J/\psi \rightarrow \mu^+\mu^-$ decay modes. Each curve spans the full physical range of its channel, $r_{max} \leq \beta^2 = 1 - 4m_l^2/M_V^2$.

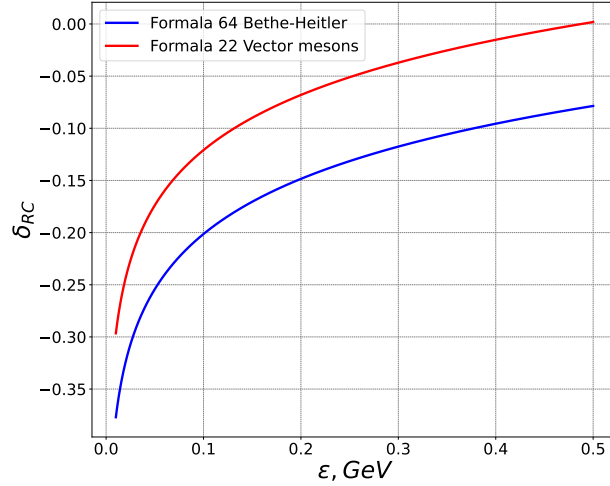


FIG. 9: Comparison of radiative corrections for Bethe Heitler process [1] with $M_{e^+e^-} = M_{J/\psi}$ and J/ψ photoproduction [4].

VIII. Invariant mass

The radiation of the hard photons affects the invariant mass $M_{e^+e^-}$ distribution in the decay $J/\psi \rightarrow e^+e^-\gamma$. The calculation of the radiation corrections depends on the maximum energy of the photon k_{max} . In the J/ψ center of mass the energy of the photon is directly connected with the $M_{e^+e^-}$ measured in the experiment

$$E_\gamma = \frac{M_{J/\psi}^2 - M_{e^+e^-}^2}{2M_{J/\psi}}. \quad (21)$$

It means that we can use cut on the $M_{e^+e^-}$ to calculate radiation corrections in Eq. 14-15 instead of the maximum energy of the radiated photon, $k_{max} = E_\gamma$. In the real experiment we study $M_{e^+e^-}$ near the J/ψ signal, $M_{e^+e^-} \sim$

$M_{J/\psi}$. In this case we can simplify the equation

$$E_\gamma = \frac{M_{J/\psi}^2 - M_{e^+e^-}^2}{2M_{J/\psi}} = \frac{(M_{J/\psi} - M_{e^+e^-})(M_{J/\psi} + M_{e^+e^-})}{2M_{J/\psi}} \approx M_{J/\psi} - M_{e^+e^-}. \quad (22)$$

It means that the parameter $k_{max} = (M_{J/\psi} - M_{e^+e^-})$ in Eq. 14-15.

IX. Summary

Acknowledgements

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